



AT | AIDTRACK

By WHIRLWIND

A Relief Distribution and Tracking System for Disaster-Affected Communities

Aldyne A. Ronquillo
Researcher

John Mark S. Salas
Researcher

Sean Austine M. Simon
Researcher

Mark Anthony D. Madalipay
Research Adviser

Grace D. Tiqui
Research Adviser

ABSTRACT

This study presents the design, development, and evaluation of AidTrack, a web-based system aimed at enhancing disaster relief operations in Santo Tomas, Pampanga. Using a mixed-methods approach, qualitative data were gathered through interviews and focus group discussions with barangay officials and MSWD personnel, while quantitative data were collected via post-test surveys using the System Usability Scale (SUS) and Technology Acceptance Model (TAM). The system was developed using the Agile Scrum methodology, enabling iterative improvements based on continuous user feedback. Key features of AidTrack include household profile management, QR code-based beneficiary verification with JWT-secured access, inventory tracking with low-stock and potential shortage alerts, relief distribution scheduling, eligibility checking, and AI-assisted prioritization

and chatbot support through Dialogflow and Gemini. Evaluation results indicate that the system is user-friendly, reliable, and contextually relevant, improving coordination, accountability, and operational efficiency. Security measures and AI integration further ensure data integrity and timely decision-making. The study concludes that AidTrack transforms manual, error-prone relief processes into a streamlined, data-driven workflow, providing local government units with a practical tool for efficient, transparent, and equitable disaster management.

Keywords: AidTrack, disaster relief management, QR code verification, inventory monitoring, AI-assisted prioritization, chatbot, Agile Scrum, system usability, Barangay operations, digital disaster management

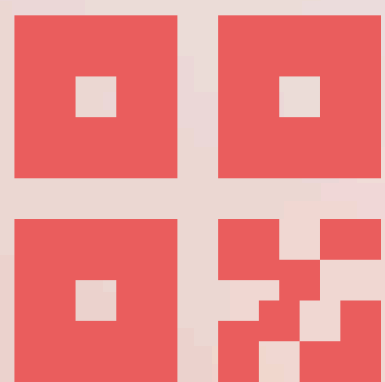
OBJECTIVES



REAL - TIME TRACKING (INVENTORY, ALERTS)



FORECASTING AND AI SUPPORT (PRIORITIZE NEEDS)



QR VERIFICATION (PREVENT DUPLICATE CLAIM)



EVALUATION (SUS, TAM STANDARDS)



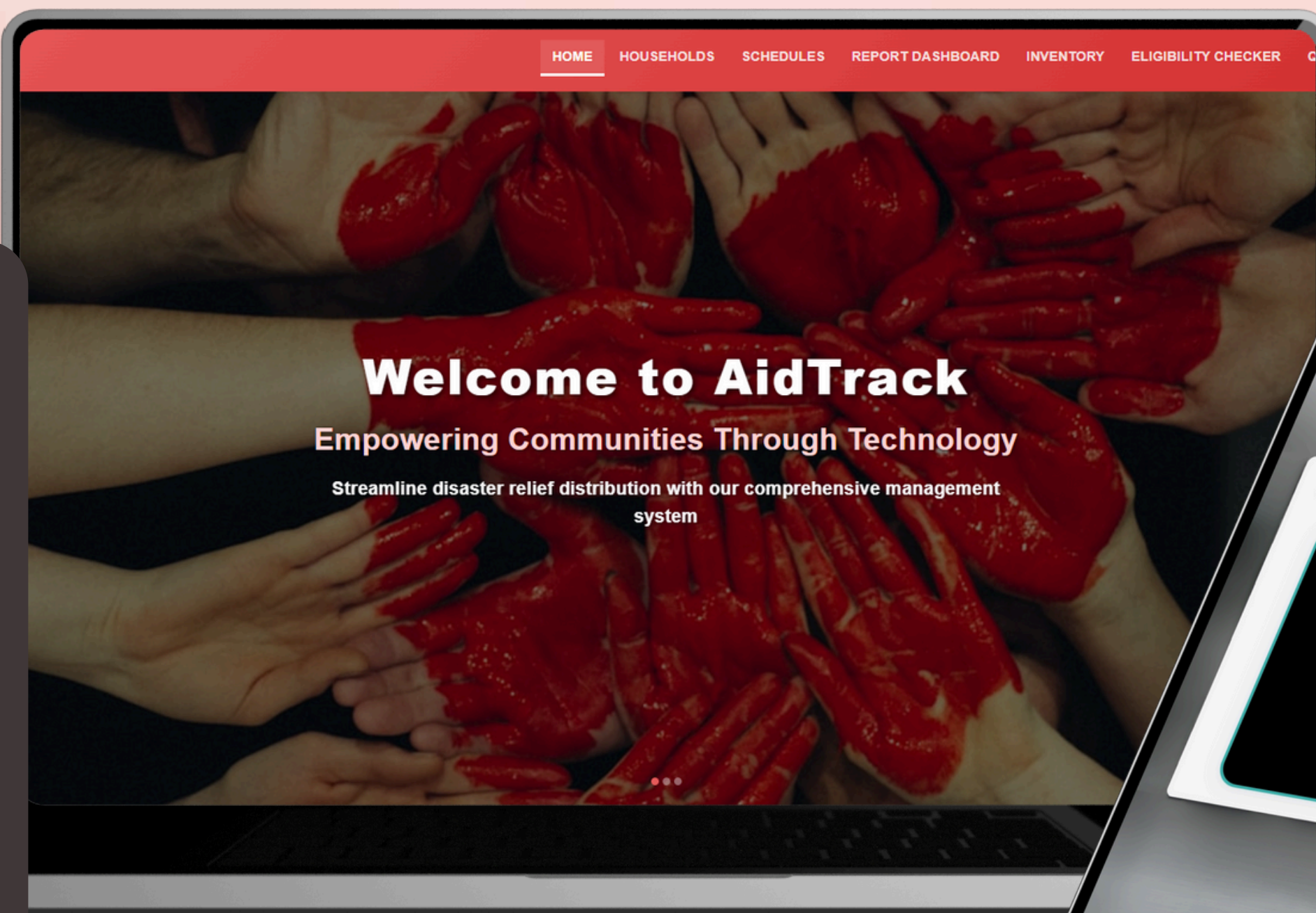
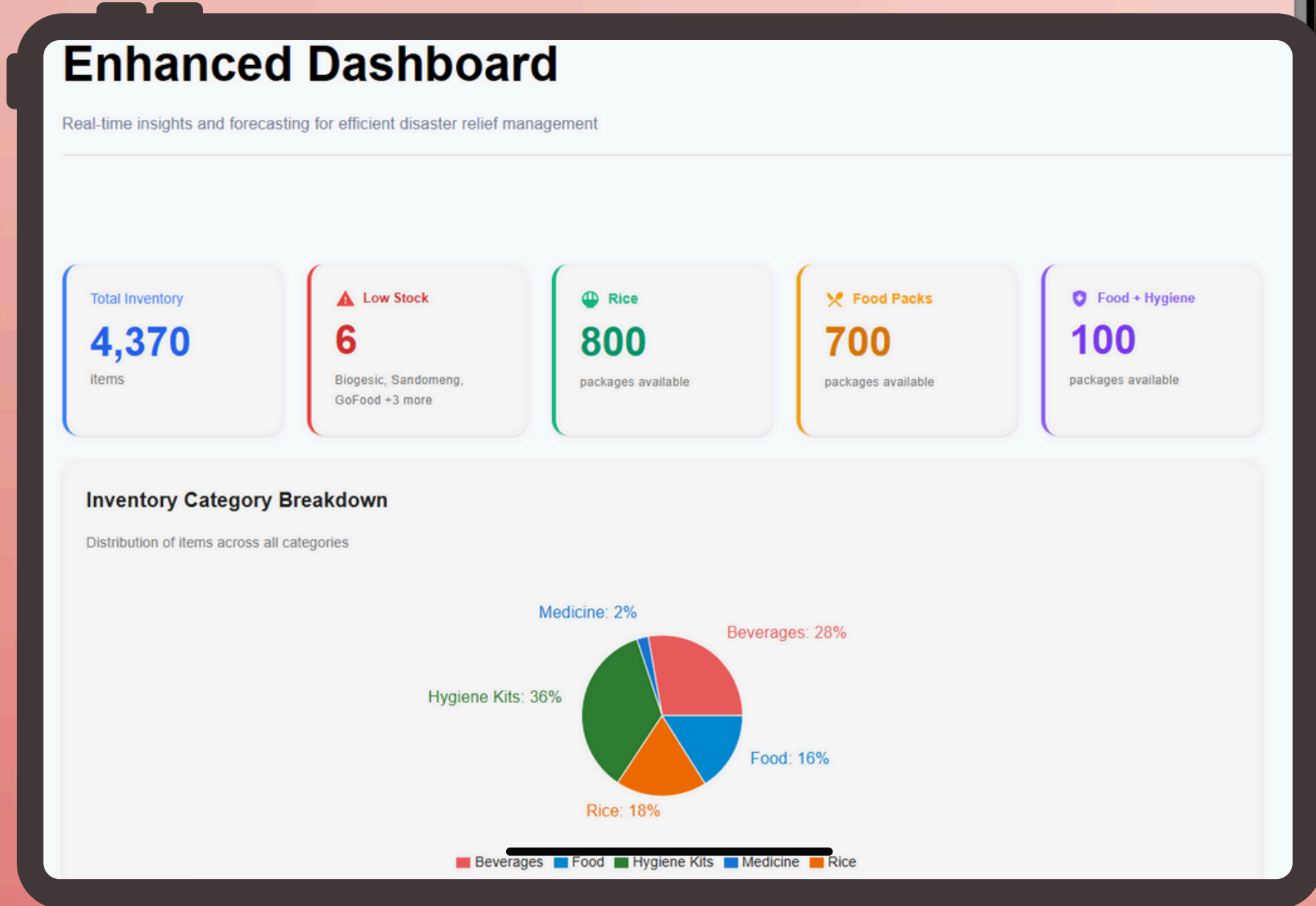
RESEARCH DESIGN MIXED METHOD (INTERVIEW, SURVEY)

TECHNOLOGIES (REACT, NODE.JS, MONGODB, DIALOGFLOW)



SOFTWARE DEVELOPMENT AGILE SCRUM

OUR SYSTEM



DISCUSSION

THE AIDTRACK SYSTEM IS DESIGNED WITH A ROBUST ARCHITECTURE AND A SUITE OF INTEGRATED FEATURES THAT COLLECTIVELY SUPPORT EFFICIENT, TRANSPARENT, AND DATA-DRIVEN DISASTER RELIEF OPERATIONS. ITS ARCHITECTURE COMBINES MODULAR COMPONENTS THAT WORK TOGETHER SEAMLESSLY, ENSURING SCALABILITY, SECURITY, AND EASE OF USE FOR BOTH ADMINISTRATORS AND END-USERS. EACH FEATURE—FROM HOUSEHOLD PROFILE MANAGEMENT AND INVENTORY MONITORING TO QR CODE GENERATION, ELIGIBILITY VERIFICATION, AND AI-POWERED ASSISTANCE—ADDRESSES SPECIFIC CHALLENGES COMMONLY ENCOUNTERED IN RELIEF DISTRIBUTION. BY PRESENTING BOTH THE OVERALL SYSTEM FRAMEWORK AND ITS KEY FUNCTIONALITIES, THIS SECTION HIGHLIGHTS HOW AIDTRACK ENHANCES COORDINATION, MINIMIZES ERRORS, AND STRENGTHENS ACCOUNTABILITY IN DELIVERING AID TO DISASTER-AFFECTED COMMUNITIES.

RESULTS

- IMPROVED EFFICIENCY, FAIRNESS, AND ACCOUNTABILITY IN RELIEF DISTRIBUTION
- QR-BASED VERIFICATION, REAL-TIME INVENTORY, AND AI FEATURES ENSURED TRANSPARENT AND ORGANIZED OPERATIONS
- VERY HIGH USABILITY, USEFULNESS, AND EASE OF LEARNING
- SYSTEM ADDRESSED REAL OPERATIONAL NEEDS
- PLATFORM SUPPORTS BETTER COORDINATION, REDUCES AID DUPLICATION, AND PROMOTES EQUITABLE DISASTER ASSISTANCE



SCAN ME!